Custom Siamese-GAN for Cats & Dogs (CIFAR-10): Similarity-Driven Adversarial Training

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Abstract—We implement a custom GAN in which the discriminator is replaced by a Siamese network that estimates the cosine similarity between a real image and a generated image. The Siamese loss encourages the model to increase the distance between real–fake pairs while keeping real–real pairs close. We train on the Cats & Dogs subset of CIFAR-10 on a T4 GPU and report loss curves, cosine-similarity histograms, nearest-neighbor qualitative checks, and image grids. Index Terms— Generative Adversarial Networks, Siamese Networks, Similarity Learning, CIFAR-10, Cats & Dogs

I. INTRODUCTION

Standard GAN discriminators output a real/fake probability. In this assignment, we reformulate the objective using a Siamese network S(x, y) that predicts a similarity score between two images. During training, S is optimized so that similar pairs (real–real) score high and dissimilar pairs (real–fake) score low, while the generator is encouraged to increase S(real, G(z)). This aligns the adversarial signal with representation similarity instead of binary classification.

II. DATASET AND PREPROCESSING

Dataset: CIFAR-10 filtered to cats and dogs, resized to 32×32 with per-channel normalization. Augmentations: random horizontal flip and slight color jitter to decorrelate batches. Batches contain both positive (real–real) and negative (real–fake) pairs for the Siamese head.

III. MODEL

Generator. DCGAN-style up sampling network producing 32×32 RGB images from latent z (dim 128). **Siamese network S.** Two weight-sharing CNN towers whose embeddings are cosine-compared; a small MLP projects to a scalar similarity. **Objective.** The Siamese loss uses margin-based contrastive terms; the generator receives a negated similarity objective so that fakes become more similar to reals. For stability we use feature-matching and EMA on generator weights.

IV. TRAINING

Optimization uses Adam (β1=0.5, β2=0.999) with equal learning rates for both networks. The loop alternates between updating S on mixed positive/negative pairs and updating G to maximize similarity. We log per-step losses for G and S and periodically export image grids and nearest-neighbor (real ↔ generated) panels.

Metric	Value
Mean RR (real-real) ↑	0.6252
Mean RF (real-fake) ↓	0.0136
GAP ↑	0.6115
FID-like proxy ↓	0.2032

Table I. Summary metrics from the Siamese embedding space.

V. RESULTS

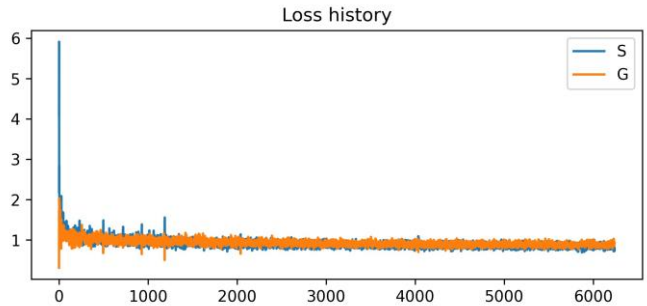

Fig. 1. Loss history for Siamese (S) and Generator (G).

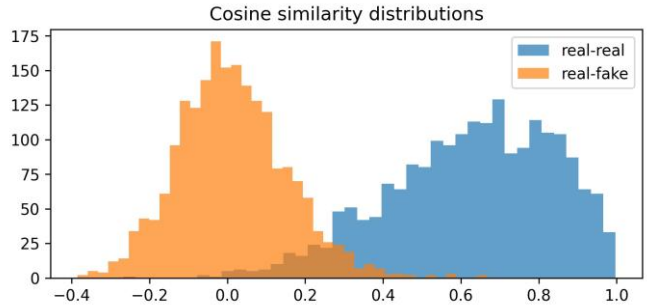

Fig. 2. Cosine similarity histograms for real–real and real–fake pairs.

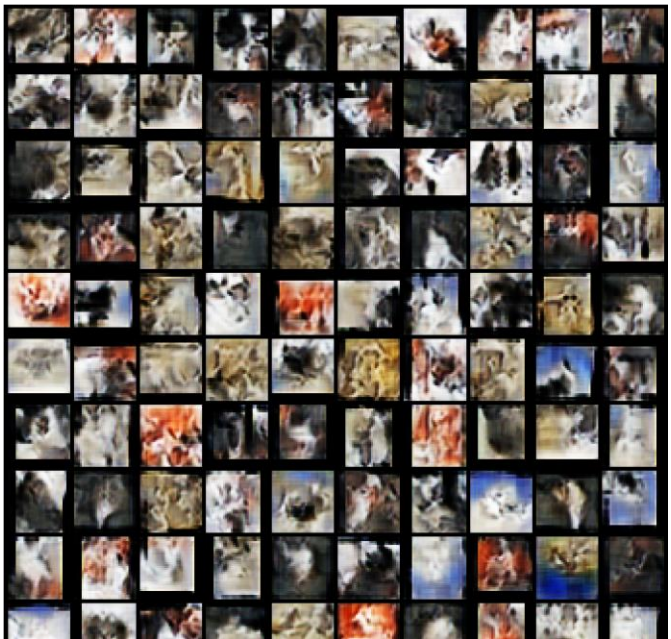

Fig. 4. Nearest-neighbor qualitative checks in the Siamese space.

Fig. 1 shows the loss history of the Siamese (S) and generator (G), which stabilizes after an initial transient. Fig. 2 overlays cosine-similarity histograms: real–real pairs cluster toward high similarity while real–fake pairs shift left, indicating the Siamese splits the two distributions. Fig. 3 presents final 10×10 sample grids; images capture coarse structure (fur textures, silhouettes) but remain blurry compared to full-resolution models. Fig. 4 shows nearest-neighbor qualitative checks in the Siamese embedding space (real anchors with their closest generated samples).

VI. DISCUSSION

The similarity-driven objective provides a stronger, semantically aligned training signal than a binary discriminator, which helps on small, two-class data. Remaining issues include texture artifacts and occasional mode bias toward dominant colors. Further improvements: (i) spectral normalization or gradient penalty for the Siamese towers, (ii) two-time-scale updates (TTUR), (iii) increasing image size to 64×64 with residual blocks, and (iv) using a CLIP-like embedding to compute similarity.

VII. CONCLUSION

A Siamese-based GAN trained on CIFAR-10 cats and dogs learns a meaningful similarity space and generates reasonable 32×32 samples. The cosine histogram gap and qualitative neighbors confirm the objective is working, though higher resolution and stronger regularization would be needed for production quality.

VIII. PROMPTS

Prompts/automation used: load CIFAR-10 cats & dogs; build pair samplers; implement Siamese towers + cosine head; define losses for real–real vs real–fake; train with Adam; log loss history; plot cosine histograms; export grids and nearest-neighbor panels; summarize metrics.

REFERENCES

[1] A. Radford et al., "Unsupervised Representation Learning with Deep Convolutional GANs," ICLR, 2016.

[3] G. Koch et al., "Siamese Neural Networks for One-Shot Image Recognition," ICML WS, 2015.

[4] M. Arjovsky et al., "Towards Principled Methods for Training GANs," 2017 (stability insights).